

1. Administrative & Study Identification

Full Study Title: An Interventional Phase 3, Double-Blind, Randomized Study to Evaluate Efficacy and Safety of PF-08634404 in Combination with Chemotherapy versus Pembrolizumab in Combination with Chemotherapy in Adult Participants with Locally Advanced or Metastatic Non-Small Cell Lung Cancer **Public Study Title:** A Study to Learn About the Study Medicine Called PF-08634404 in Combination With Chemotherapy in Adult Participants With Locally Advanced or Metastatic Non-Small Cell Lung Cancer **Short Title/Acronym:** Symbiotic-Lung-01 **Protocol Number:** C6461001 **NCT Number:** NCT07222566 **Posting ID:** 73445812145816655 **Sponsor/Company Name:** Pfizer **Investigational Product:** PF-08634404 (Anti-PD-1 x VEGF bispecific antibody) **Phase of Development:** Phase 3 (PHASE3) **Trial Status:** NOT_YET_RECRUITING (Not yet recruiting participants) **Medical Monitor:** Pfizer CT.gov Call Center / Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate if the investigational bispecific antibody PF-08634404, when given in combination with chemotherapy, is more effective than the current standard of care (pembrolizumab combined with chemotherapy) in treating adults with a type of lung cancer called non-small cell lung cancer (NSCLC) that is either locally advanced or has spread to other parts of the body. **Secondary Objectives:** To assess Objective Response Rate (ORR), Progression-Free Survival (PFS) as assessed by the investigator, and the overall safety and tolerability profile of PF-08634404 compared to the pembrolizumab regimen.

2.2 Background & Rationale Despite recent advancements in treatment options, the prognosis for patients with advanced NSCLC remains very challenging. PF-08634404 is an investigational bispecific antibody that targets both PD-1 and VEGF pathways simultaneously, utilizing a tetravalent structure designed to modulate VEGF-related immune pathways and enhance PD-1 binding. It is hypothesized that dual targeting of these proteins may help the immune system better identify and attack cancer cells while slowing tumor vascular growth, potentially offering superior efficacy compared to targeting PD-1 alone.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Pembrolizumab + Chemotherapy) **Blinding:** Double-blind **Randomization:** Randomized (1:1 to Experimental or Control groups)

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 1,500 evaluable participants **Number of Sites:** Global Multicenter **Estimated Study Start:** October 2025 **Estimated Primary Completion:** 27-Feb-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years at screening. 2. Pathologically confirmed locally advanced (Stage IIIB/IIIC) or metastatic (Stage IV) squamous or non-squamous NSCLC (according to the 9th edition of the TNM staging system). 3. Not a candidate for complete surgical resection and curative concurrent/sequential chemoradiotherapy. 4. Must be treatment-naïve for advanced or metastatic disease (no prior systemic anti-tumor therapy). 5. Have tumor tissue available (paraffin block or slides) and PD-L1 status available based on local testing results. 6. Measurable disease based on RECIST v1.1 per investigator. 7. Eastern Cooperative Oncology Group (ECOG) performance status score of 0 or 1. 8. Expected survival ≥ 12 weeks.

4.2 Exclusion Criteria 1. Known actionable genomic alterations (AGAs), including EGFR, ALK, ROS1, NTRK, BRAF, RET, and MET. Documented negative results for EGFR, ALK, and ROS1 are required for non-squamous histology. 2. Prior treatment with immunotherapy or systemic anti-angiogenic therapy. 3. Known active CNS lesions (definitively treated brain metastases or clinically inactive brain metastases < 1 cm are permitted). 4. Evidence of noninfectious or drug-induced interstitial lung disease (ILD) or pneumonitis, or Grade ≥ 3 pulmonary disease unrelated to malignancy. 5. Clinically significant risk of hemorrhage or fistula. 6. Any history of another malignancy within 3 years before the first dose, or evidence of residual disease. 7. Unresolved toxicities from prior anti-tumor therapy that did not recover to NCI CTCAE v5.0 Grade 0 or 1. 8. History of severe allergy to study interventions, or history of severe allergic reaction to a chimeric or humanized antibody. 9. History of allogeneic organ / hematopoietic stem cell transplantation. 10. Acute, chronic, or symptomatic infections including active HIV, HBV, or HCV.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants are stratified by histology (Part 1 for squamous, Part 2 for non-squamous). * **Experimental Arm (Part 1 - Arm A or Part 2 - Arm C):** PF-08634404 + Chemotherapy (regimen specific to tumor type) followed by maintenance with PF-08634404 $\pm$ chemotherapy. * **Control Arm (Part 1 - Arm B or Part 2 - Arm D):** Pembrolizumab (Trade Name: Keytruda | ATC Code: L01FF02 | EMA URL: https://www.ema.europa.eu/en/documents/product-information/keytruda-epar-product-information_en.pdf) + Chemotherapy (regimen specific to tumor type) followed by maintenance with Pembrolizumab $\pm$ chemotherapy. **Route of Administration:** Intravenous (IV) infusion **Duration of Treatment:** Administered in cycles until disease progression, unacceptable toxicity, or investigator decision.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care. **Prohibited:** Therapeutic oral or parenteral anticoagulants or thrombolytic agents (< 10 days prior to first dose), chronic antiplatelet therapy (< 7 days prior), live or attenuated live vaccines (< 4 weeks prior), current high-dose systemic corticosteroids, and prohibited concomitant medication(s) < 21 days to the first dose.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor/Genomic testing, baseline imaging
Treatment Cycles	Every cycle	IV infusion of study drug + chemo, Safety Labs, PK/ADA collection
Tumor Assessment	Weeks 6 to 48	Imaging/scans to monitor tumor response every 6 weeks
Long-term Assessment	$\geq$ Week 48	Imaging/scans every 12 weeks thereafter
End of Study	Until PD/Toxicity	Discontinuation visit, final safety check, survival follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Overall Survival (OS):** Defined as the time from the date of randomization to the date of death due to any cause. 2. **Progression-Free Survival (PFS) assessed by Blinded Independent Central Review (BICR):** Defined as the time from the date of randomization to the date of the first documentation of objective progressive disease (PD) assessed by BICR per RECIST v1.1, or death due to any cause, whichever occurs first.

7.2 Secondary Endpoints 1. Confirmed objective response rate (ORR) using RECIST v1.1 as assessed by BICR. 2. Progression-Free Survival (PFS) as assessed by Investigator. 3. Confirmed ORR using RECIST v1.1 as assessed by investigator.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring at each clinical visit including specific screening for immune-related adverse events and VEGF-related toxicities (e.g., hemorrhage, hypertension). **Serious Adverse Events (SAEs):** Reported standardly within 24 hours. Participants with significant risk of hemorrhage or fistula are excluded and monitored closely if risk develops. **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee oversees unblinded safety and efficacy data across global sites.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all primary efficacy endpoints. Safety Analysis Set for participants receiving ≥ 1 dose of study drug. **Sample Size Justification:** $N = 1,500$ participants is adequately powered to detect statistically significant improvements in PFS/OS for the PF-08634404 group compared to the pembrolizumab active control group. **Handling of Missing Data:** Standard Kaplan-Meier methods for time-to-event endpoints; patients without documented progression or death will be right-censored at the last evaluable tumor assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Criteria Requirement:** Trial must be in PHASE3 (criteria_type: PHASE_REQUIREMENT, phase_id: PHASE3). **Monitoring Plan:** Routine onsite and remote monitoring to ensure protocol adherence, drug accountability, and accurate safety reporting. **Audit Trail:** Validated eCRF systems, robust query resolution protocols, and independent auditing of RECIST measurements via central review.

11. Study Identifiers & Governance

Primary ID: {{C6461001}} **Registry Number:** {{NCT07222566}} **Posting ID:** {{73445812145816655}} **Sponsor/Lead Organization:** {{Pfizer}} **Study Title:** {{Symbiotic-Lung-01}} **Trial Status:** {{NOT_YET_RECRUITING}}

12. Clinical Framework & Disease Ontologies

Primary Condition / Preferred UMLS Name: {{Advanced Non-Small Cell Lung Cancer}} **Disease Areas:** {{Advanced Non-Small Cell Lung Cancer, Non-Small Cell Lung Cancer, Carcinoma, Non-Small-Cell

Lung, Carcinoma, Non-Small-Cell Lung (NSCLC), Metastatic Non Small Cell Lung Cancer, Lung Cancer}} **Disease Entity / Preferred Name:** {{Carcinoma}} **Semantic Type:** {{Neoplastic Process}} **Definition:** {{A malignant tumor arising from epithelial cells. Carcinomas that arise from glandular epithelium are called adenocarcinomas, those that arise from squamous epithelium are called squamous cell carcinomas, and those that arise from transitional epithelium are called transitional cell carcinomas (NCI Thesaurus). [<https://orcid.org/0000-0002-0736-9199>]}} **MeSH Code:** {{D002277}} **HPO Code:** {{HP:0030731}} **SNOMED ID:** {{1187425009}} **SNOMED Hierarchy:** {{1187225007|Malignant epithelial neoplasm; 118285006|Epithelial neoplasm; 108369006|Neoplasm; 1240414004|Malignant neoplasm}} **Source Level:** {{5}}

13. Study Procedures & Methodology

Management Pathway: {{Standard Oncology Care Pathway for advanced NSCLC}} **Education Protocol (HCP):** {{Protocol training, RECIST v1.1 assessment, and immune-mediated AE management}} **Education Protocol (Patient):** {{Informed consent, infusion schedule understanding, and AE reporting education}} **Quality Audit Mechanism:** {{Blinded Independent Central Review (BICR) for imaging & continuous site monitoring}}

14. Observation & Follow-Up Schedule

Total Duration: {{Until disease progression, unacceptable toxicity, or investigator decision}} **Onsite Visit Cadence:** {{Every infusion cycle}} **Remote Monitoring Frequency:** {{As needed per protocol for survival follow-up}} **Checklist Review Interval:** {{Imaging every 6 weeks for the first 48 weeks, then Q12W}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				